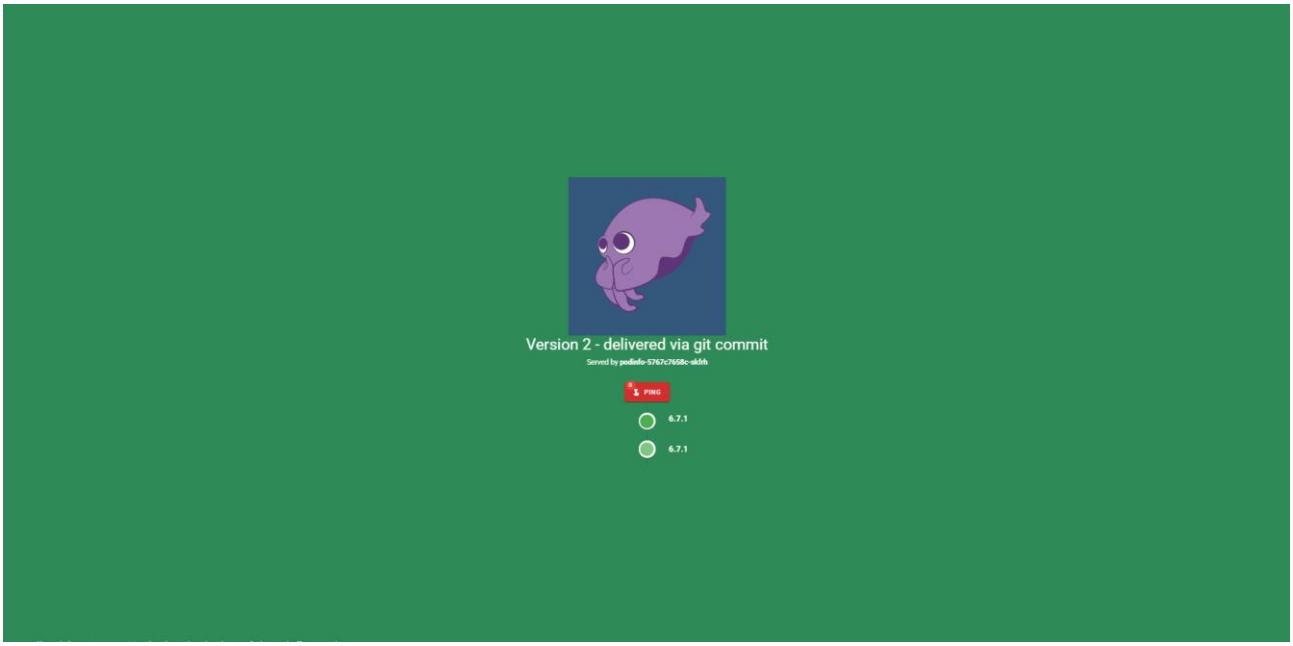
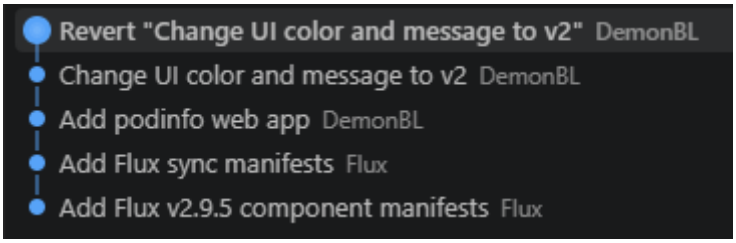



```
PS F:\trenee\concourse-lab> fly -t local trigger-job -j hello-world/hello-world
started hello-world/hello-world #9
PS F:\trenee\concourse-lab> fly -t local watch -j hello-world/hello-world
initializing
initializing check: image
selected worker: local-worker
selected worker: local-worker
fetching bash@sha256:61962062d969cb46dfc2bad061d36342406fa485f64f246aa7e95693ca07df1f
e2de96513ba9 [-------------------------------------------] 0.0b/3.7MiB
2230b27b856a [-------------------------------------------] 0.0b/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [=====>-------------------------------------] 506.3KiB/3.7MiB
2230b27b856a [-------------------------------------------] 0.0b/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [=============>-----------------------------] 1.2MiB/3.7MiB
2230b27b856a [-------------------------------------------] 0.0b/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==================================>----------] 2.9MiB/3.7MiB
2230b27b856a [-------------------------------------------] 0.0b/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [-------------------------------------------] 0.0b/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [=>-----------------------------------------] 136.0KiB/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [=========================>-----------------] 1.7MiB/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [====================================>-------] 2.3MiB/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [==========================================] 2.8MiB/2.8MiB
a4675d78db22 [-------------------------------------------] 0.0b/337.0b
e2de96513ba9 [==========================================] 3.7MiB/3.7MiB
2230b27b856a [==========================================] 2.8MiB/2.8MiB
a4675d78db22 [==========================================] 337.0b/337.0b
selected worker: local-worker
running bash -c echo "hello world"

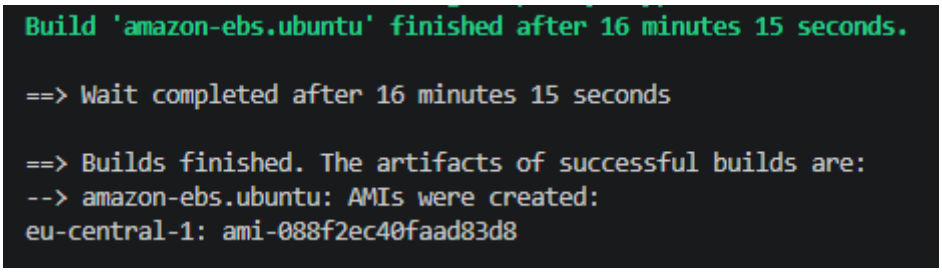
hello world
succeeded
```

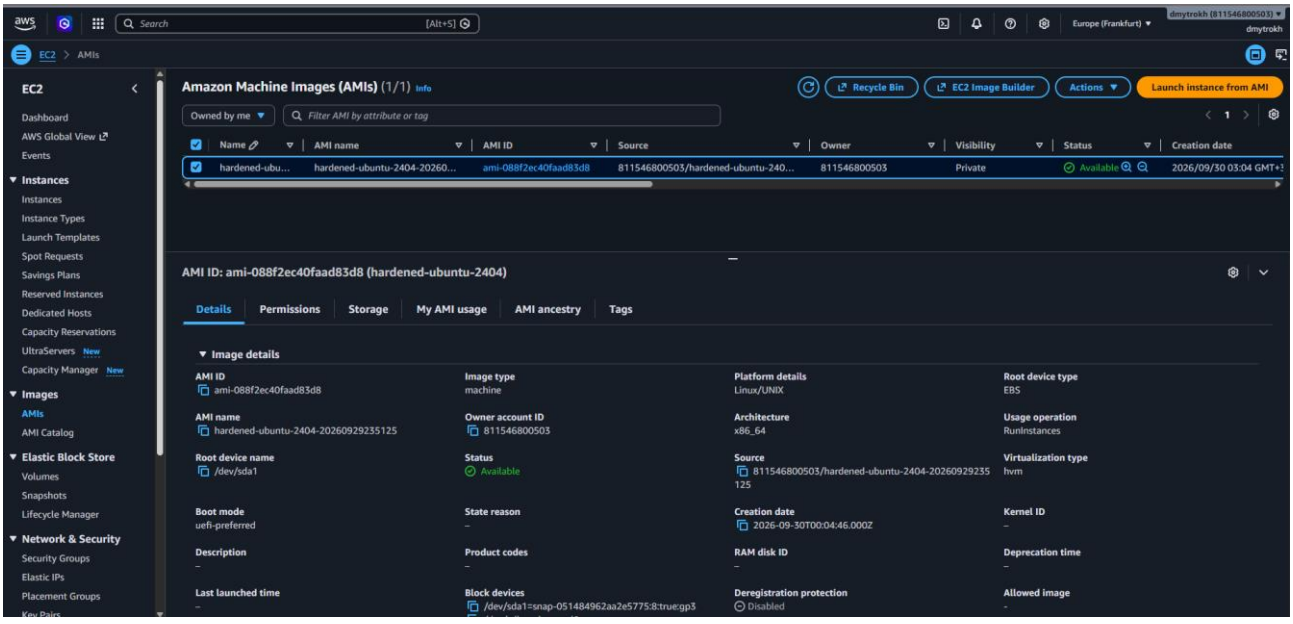
Task2


Version 2 - delivered via git commit
PING
6.7.1
6.7.1


Revert "Change UI color and message to v2" DemonBL
Change UI color and message to v2 DemonBL
Add podinfo web app DemonBL
Add Flux sync manifests Flux
Add Flux v2.9.5 component manifests Flux

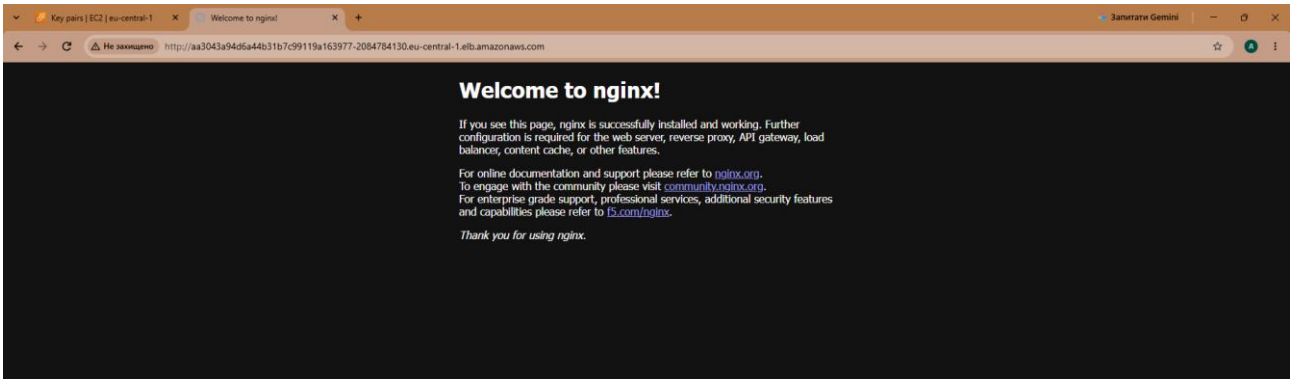
Task3


Build 'amazon-ebs.ubuntu' finished after 16 minutes 15 seconds.
==> Wait completed after 16 minutes 15 seconds
==> Builds finished. The artifacts of successful builds are:
--> amazon-ebs.ubuntu: AMIs were created:
eu-central-1: ami-088f2ec40faad83d8


Amazon Machine Images (AMIs) (1/1)
Recycle Bin
EC2 Image Builder
Actions
Launch instance from AMI
hardened-ubu...
hardened-ubuntu-2404-20260...
ami-088f2ec40faad83d8
811546800503/hardened-ubuntu-240...
811546800503
Private
Available
2026/09/30 03:04 GMT+3
AMI ID: ami-088f2ec40faad83d8 (hardened-ubuntu-2404)
Details
Permissions
Storage
My AMI usage
AMI ancestry
Tags
Image details
AMI ID
ami-088f2ec40faad83d8
Image type
machine
Platform details
Linux/UNIX
Root device type
EBS
AMI name
hardened-ubuntu-2404-20260929235125
Owner account ID
811546800503
Architecture
x86_64
Usage operation
RunInstances
Root device name
/dev/sda1
Status
Available
Source
811546800503/hardened-ubuntu-2404-20260929235125
Virtualization type
hvm
Boot mode
uefi-preferred
State reason
Creation date
2026-09-30T00:04:46.000Z
Kernel ID
Description
Product codes
RAM disk ID
Deprecation time
Last launched time
Block devices
/dev/sda1=snap-05148496 2aa2e5775:8:true:gp3
Deregistration protection
Disabled
Allowed image

```
ubuntu@ip-172-31-11-180:~$ systemctl is-active auditd
active
ubuntu@ip-172-31-11-180:~$ sysctl kernel.randomize_va_space net.ipv4.conf.all.accept_redirects net.ipv4.conf.all.send_redirects
kernel.randomize_va_space = 2
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
ubuntu@ip-172-31-11-180:~$ grep -E "^PASS_MAX_DAYS|^UMASK" /etc/login.defs
UMASK           027
PASS_MAX_DAYS   60
ubuntu@ip-172-31-11-180:~$ ls -l /etc/modprobe.d/ | grep dev-sec
-rw-r--r-- 1 root root  318 Sep 29 23:56 dev-sec.conf
ubuntu@ip-172-31-11-180:~$
```

Task 4




```
PS F:\trenee\eks-lab> terraform apply tfplan

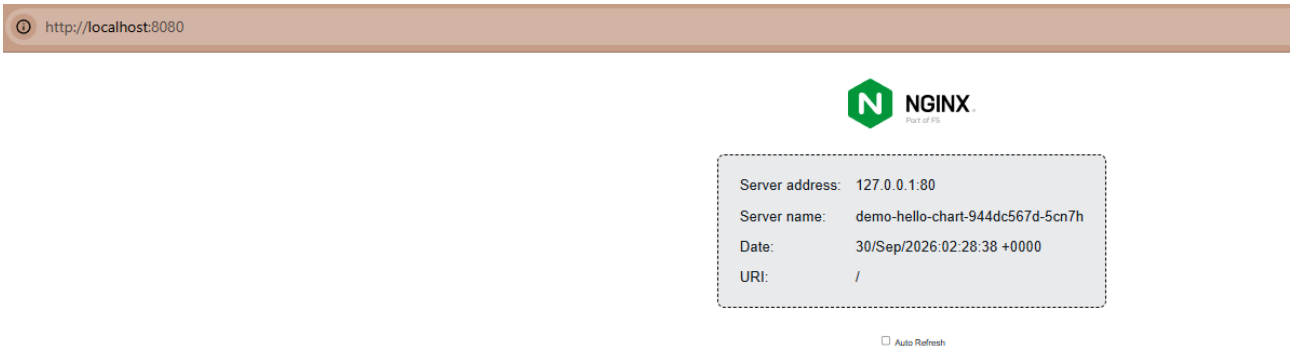
Apply complete! Resources: 3 added, 0 changed, 1 destroyed.

Outputs:

cluster_endpoint = "https://D926DDA97531C08FB3F543AFFCB5302C.gr7.eu-central-1.eks.amazonaws.com"
cluster_name = "lab-eks"
configure_kubectl = "aws eks update-kubeconfig --region eu-central-1 --name lab-eks"
PS F:\trenee\eks-lab> terraform output
cluster_endpoint = "https://D926DDA97531C08FB3F543AFFCB5302C.gr7.eu-central-1.eks.amazonaws.com"
cluster_name = "lab-eks"
configure_kubectl = "aws eks update-kubeconfig --region eu-central-1 --name lab-eks"
PS F:\trenee\eks-lab> aws eks list-nodegroups --cluster-name lab-eks
{
    "nodegroups": [
        "default-aaaf24dbb3ca63b785f7d3698e"
    ]
}

PS F:\trenee\eks-lab> kubectl get nodes -o wide
NAME                                         STATUS   ROLES    AGE   VERSION               INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                        KERNEL-VERSION                  CONTAINER-RUNTI
ME
ip-10-0-1-78.eu-central-1.compute.internal   Ready    <none>   8m    v1.34.11-eks-f4fc4f1   10.0.1.78     <none>        Amazon Linux 2023.12.20260918   6.12.103-129.197.amzn2023.x86_64   containerd://2.
2.7+unknown
ip-10-0-2-198.eu-central-1.compute.internal   Ready    <none>   8m    v1.34.11-eks-f4fc4f1   10.0.2.198    <none>        Amazon Linux 2023.12.20260918   6.12.103-129.197.amzn2023.x86_64   containerd://2.
2.7+unknown
PS F:\trenee\eks-lab> kubectl get pods -A
NAMESPACE     NAME                           READY   STATUS    RESTARTS   AGE
kube-system   aws-node-5r4fk                 2/2     Running   0          8m10s
kube-system   aws-node-xb7f4                 2/2     Running   0          8m10s
kube-system   coredns-57b98d58c8-4s9k2       1/1     Running   0          7m16s
kube-system   coredns-57b98d58c8-fqkwr       1/1     Running   0          7m16s
kube-system   eks-pod-identity-agent-cp9zp   1/1     Running   0          8m10s
kube-system   eks-pod-identity-agent-cw8bj   1/1     Running   0          8m10s
kube-system   kube-proxy-nb45c               1/1     Running   0          7m17s
kube-system   kube-proxy-xxgpl               1/1     Running   0          7m17s
PS F:\trenee\eks-lab> kubectl create deployment nginx --image=nginx --replicas=2
deployment.apps/nginx created
PS F:\trenee\eks-lab> kubectl expose deployment nginx --port=80 --type=LoadBalancer
service/nginx exposed
PS F:\trenee\eks-lab> kubectl get pods -o wide
NAME                     READY   STATUS    RESTARTS   AGE   IP           NODE                                         NOMINATED NODE   READINESS GATES
nginx-66686b6766-s6jdf   1/1     Running   0          16s   10.0.1.7     ip-10-0-1-78.eu-central-1.compute.internal   <none>           <none>
nginx-66686b6766-wc851   1/1     Running   0          16s   10.0.2.176   ip-10-0-2-198.eu-central-1.compute.internal   <none>           <none>
PS F:\trenee\eks-lab> kubectl get svc nginx -w
NAME    TYPE           CLUSTER-IP       EXTERNAL-IP                                                               PORT(S)        AGE
nginx   LoadBalancer   172.20.232.133   aa3043a94d6a44b31b7c99119a163977-2084784130.eu-central-1.elb.amazonaws.com   80:32617/TCP   19s
```

Task 5



```
PS F:\trenee> helm lint .\hello-chart
==> Linting .\hello-chart
[INFO] Chart.yaml: icon is recommended

1 chart(s) linted, 0 chart(s) failed
PS F:\trenee> helm list
NAME    NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                   APP VERSION
demo    default         4               2026-09-30 05:25:39.4060041 +0300 EEST  deployed        hello-chart-0.1.0       latest
PS F:\trenee> kubectl get pods --show-labels
NAME                               READY   STATUS    RESTARTS   AGE     LABELS
demo-hello-chart-944dc567d-279sv   1/1     Running   0          5m20s   app.kubernetes.io/instance=demo,app.kubernetes.io/managed-by=Helm,app.kubernetes.io/name=hello-chart,app.kubernetes.io/version=late
st,environment=prod,helm.sh/chart=hello-chart-0.1.0,pod-template-hash=944dc567d,team=devops,tier=frontend
demo-hello-chart-944dc567d-5cn7h   1/1     Running   0          5m27s   app.kubernetes.io/instance=demo,app.kubernetes.io/managed-by=Helm,app.kubernetes.io/name=hello-chart,app.kubernetes.io/version=late
st,environment=prod,helm.sh/chart=hello-chart-0.1.0,pod-template-hash=944dc567d,team=devops,tier=frontend
PS F:\trenee> helm upgrade demo .\hello-chart --set replicaCount=3 --set image.tag=plain-text
Release "demo" has been upgraded. Happy Helming!
NAME: demo
LAST DEPLOYED: Wed Sep 30 05:31:14 2026
NAMESPACE: default
STATUS: deployed
REVISION: 5
DESCRIPTION: Upgrade complete
TEST SUITE: None
NOTES:
Release "demo" of chart hello-chart deployed to namespace "default".
Image: nginxdemos/hello:plain-text
Replicas: 3

Access the app:
  kubectl port-forward svc/demo-hello-chart 8080:80 -n default
  Then open http://localhost:8080
PS F:\trenee> kubectl get deploy demo-hello-chart -o wide
NAME              READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES                       SELECTOR
demo-hello-chart   2/3     1            2           10m   hello        nginxdemos/hello:plain-text   app.kubernetes.io/instance=demo,app.kubernetes.io/name=hello-chart
PS F:\trenee> kubectl get deploy demo-hello-chart -o wide
NAME              READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES                       SELECTOR
demo-hello-chart   3/3     3            3           10m   hello        nginxdemos/hello:plain-text   app.kubernetes.io/instance=demo,app.kubernetes.io/name=hello-chart
```